Multivariate Calculus for Machine Learning - Complete Notes

1. What is Multivariate Calculus?
Studies functions with more than one variable, e.g. f(x,y) or f(x1,x2,...,xn). ML models have many features and parameters, so multivariate calculus is essential.

2. Why is it important in ML?
Used to optimize loss functions, compute gradients, train neural networks, perform gradient descent, and backpropagation.

3. Variables
Independent Variables: Inputs/features.
Dependent Variable: Output/target.

4. Function of Multiple Variables
Example: $f(x,y)=x^2+y^2$.

5. Partial Derivative
Measures change in one variable while keeping others constant.
Notation: $\partial f/\partial x$, $\partial f/\partial y$.

6. Gradient
Vector of all partial derivatives. Indicates the direction of maximum increase. Negative gradient gives the steepest descent direction.

7. Chain Rule
Used when one function depends on another. Essential for backpropagation in deep learning.

8. Gradient Descent (GD)
Optimization algorithm:
1. Initialize weights.
2. Compute loss.
3. Compute gradients.
4. Update weights.
5. Repeat until convergence.

9. Learning Rate
Controls update size. Too small = slow learning. Too large = overshooting.

10. Local vs Global Minimum
Local Minimum: Lowest nearby point.
Global Minimum: Lowest point in the entire function.

11. Hessian Matrix
Matrix of second-order partial derivatives. Used to analyze curvature and identify minima/maxima.

12. Jacobian Matrix
Matrix of first-order partial derivatives of vector-valued functions.

13. Taylor Series
Approximates complex functions using polynomials. Useful in optimization.

14. Convex vs Non-Convex Functions
Convex: One global minimum; easier to optimize.

Non-convex: Multiple local minima; common in deep learning.

15. Loss Functions
Examples: MSE (Mean Squared Error), MAE (Mean Absolute Error), Cross-Entropy Loss.

16. Applications
Linear Regression, Logistic Regression, Neural Networks, Deep Learning, SVMs, Recommendation Systems.

Interview Questions
What is a gradient? Why are partial derivatives important? What is the Chain Rule? Difference between derivative and gradient? What are Hessian and Jacobian matrices?

Acronyms
ML = Machine Learning
DL = Deep Learning
GD = Gradient Descent
MSE = Mean Squared Error
MAE = Mean Absolute Error
SVM = Support Vector Machine

Quick Revision
Partial Derivative -> One variable changes.
Gradient -> Vector of partial derivatives.
Negative Gradient -> Direction of steepest decrease.
Chain Rule -> Backpropagation.
Gradient Descent -> Minimizes loss.
Learning Rate -> Controls update size.
Hessian -> Second derivatives.
Jacobian -> First derivatives.